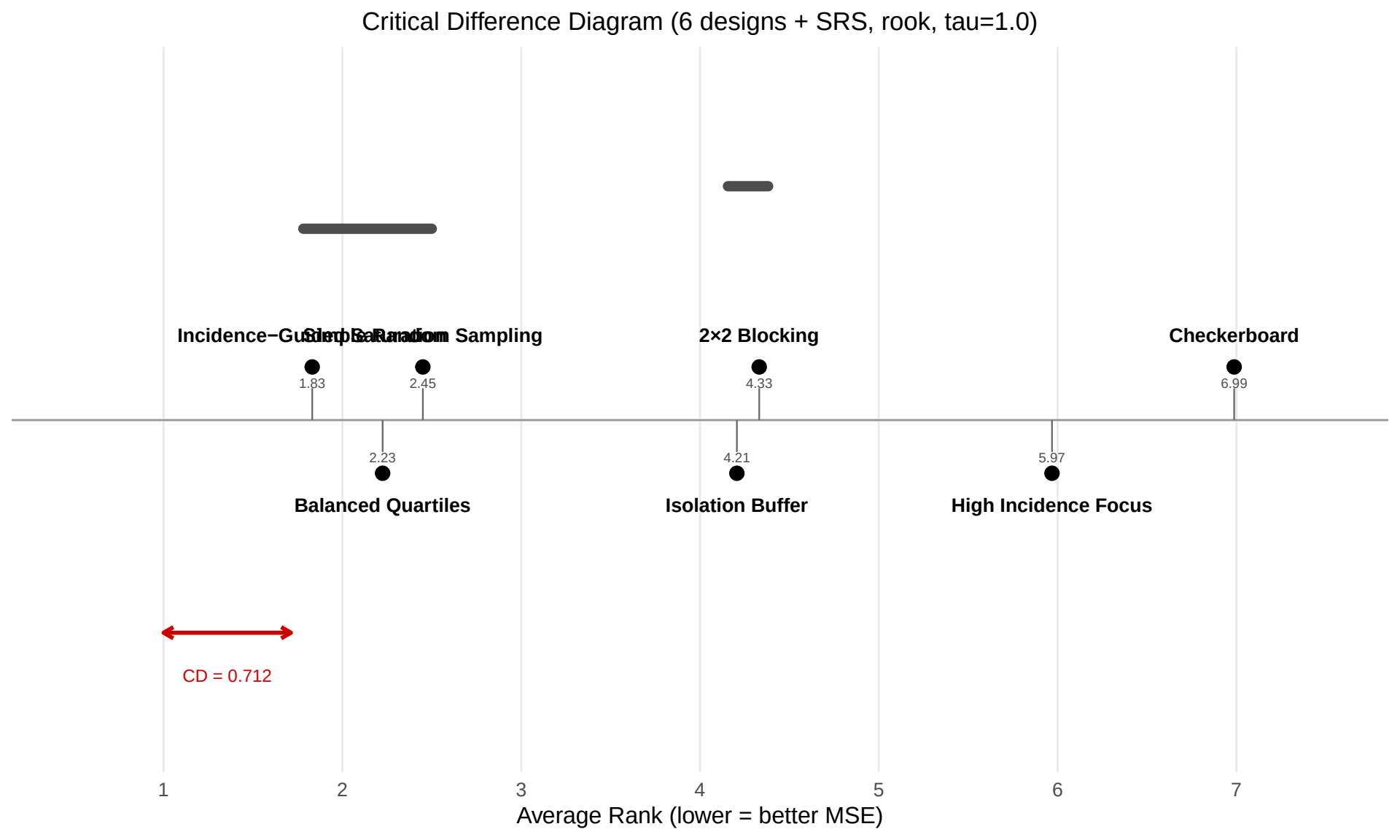
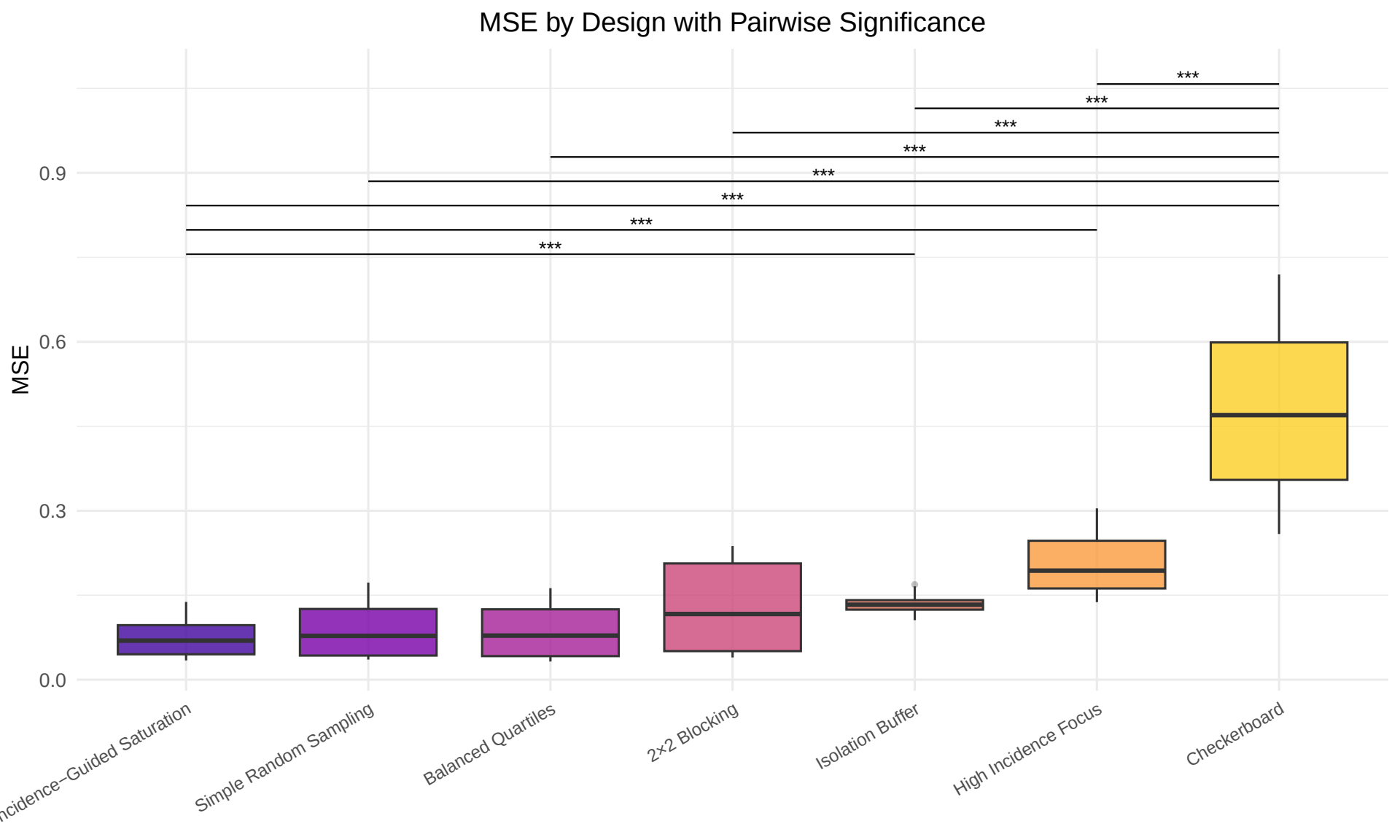
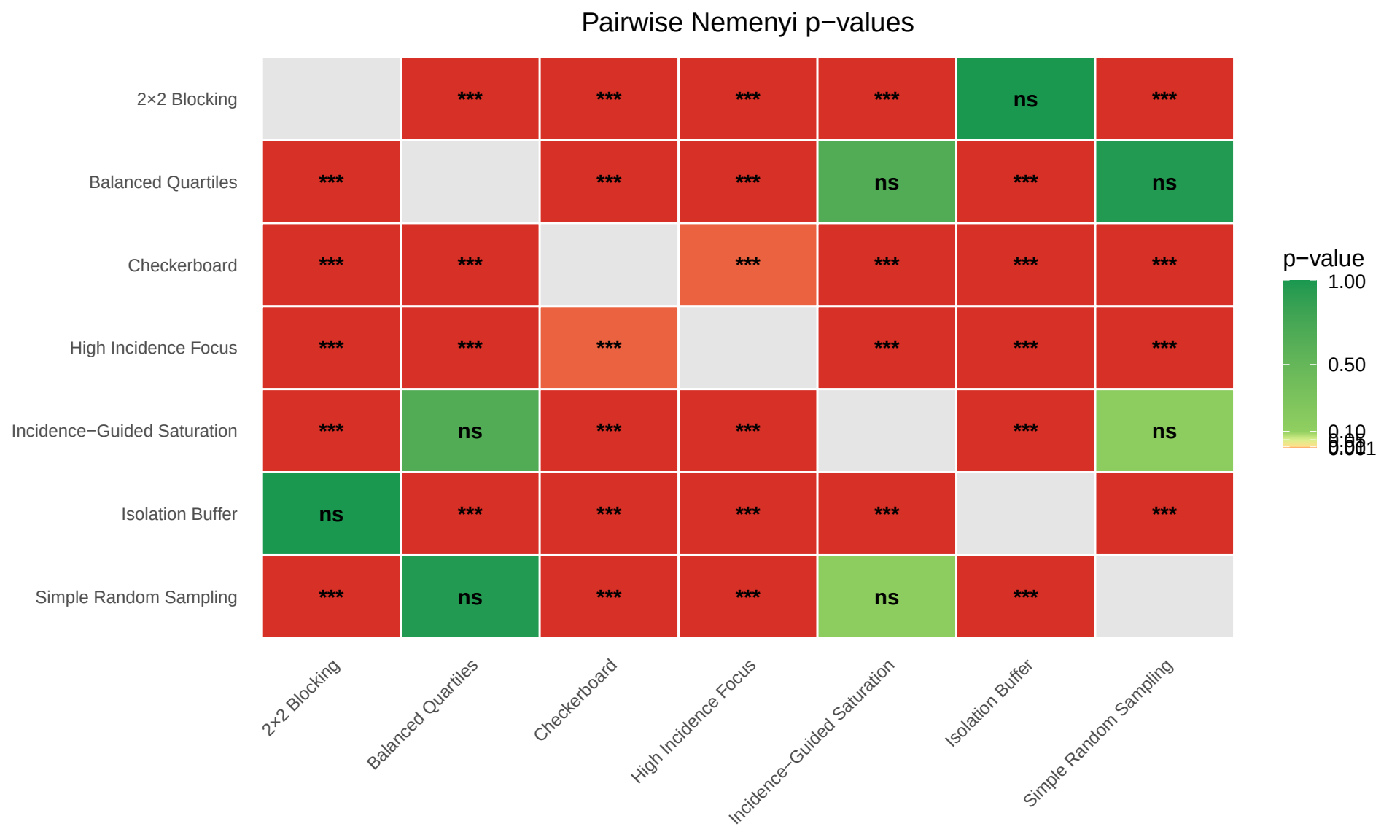
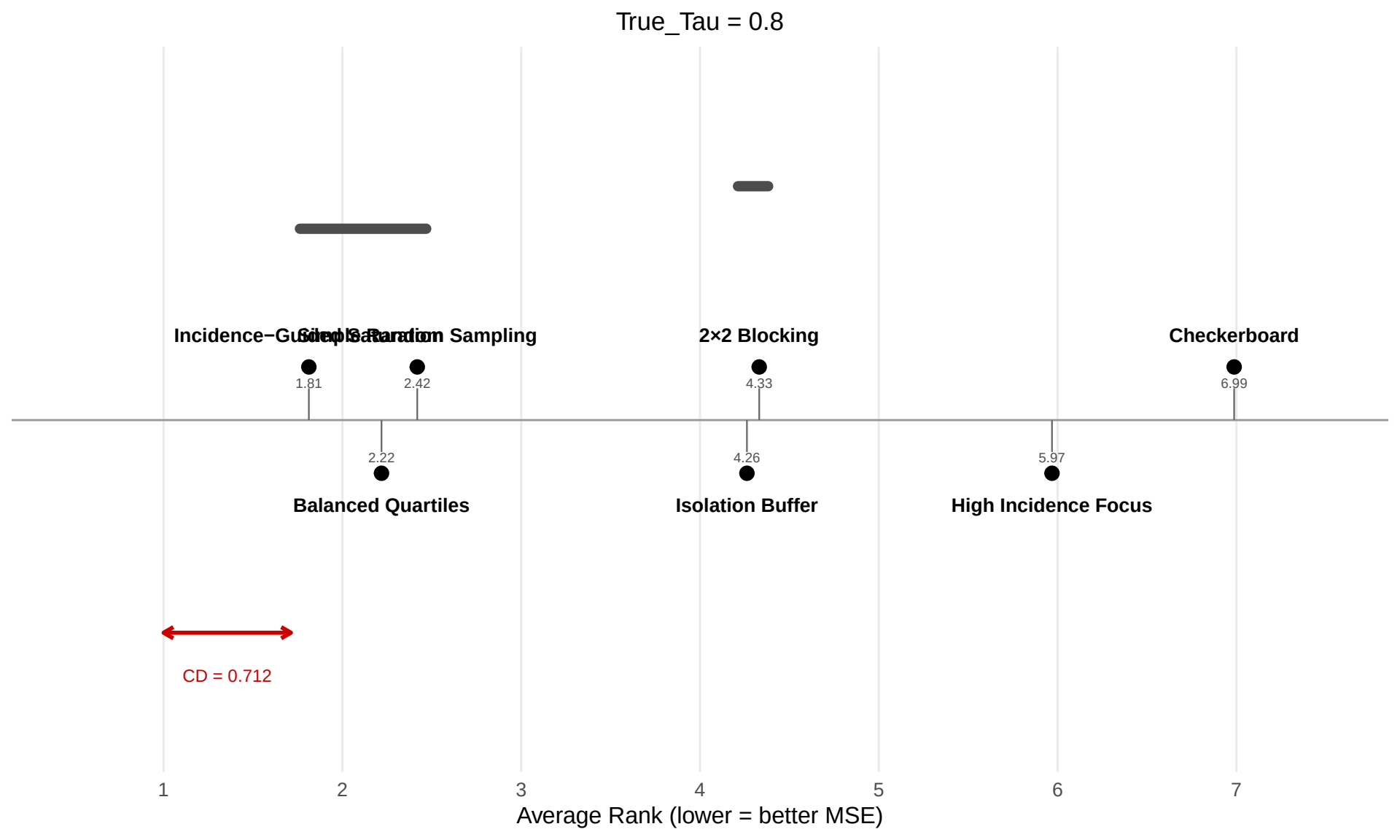
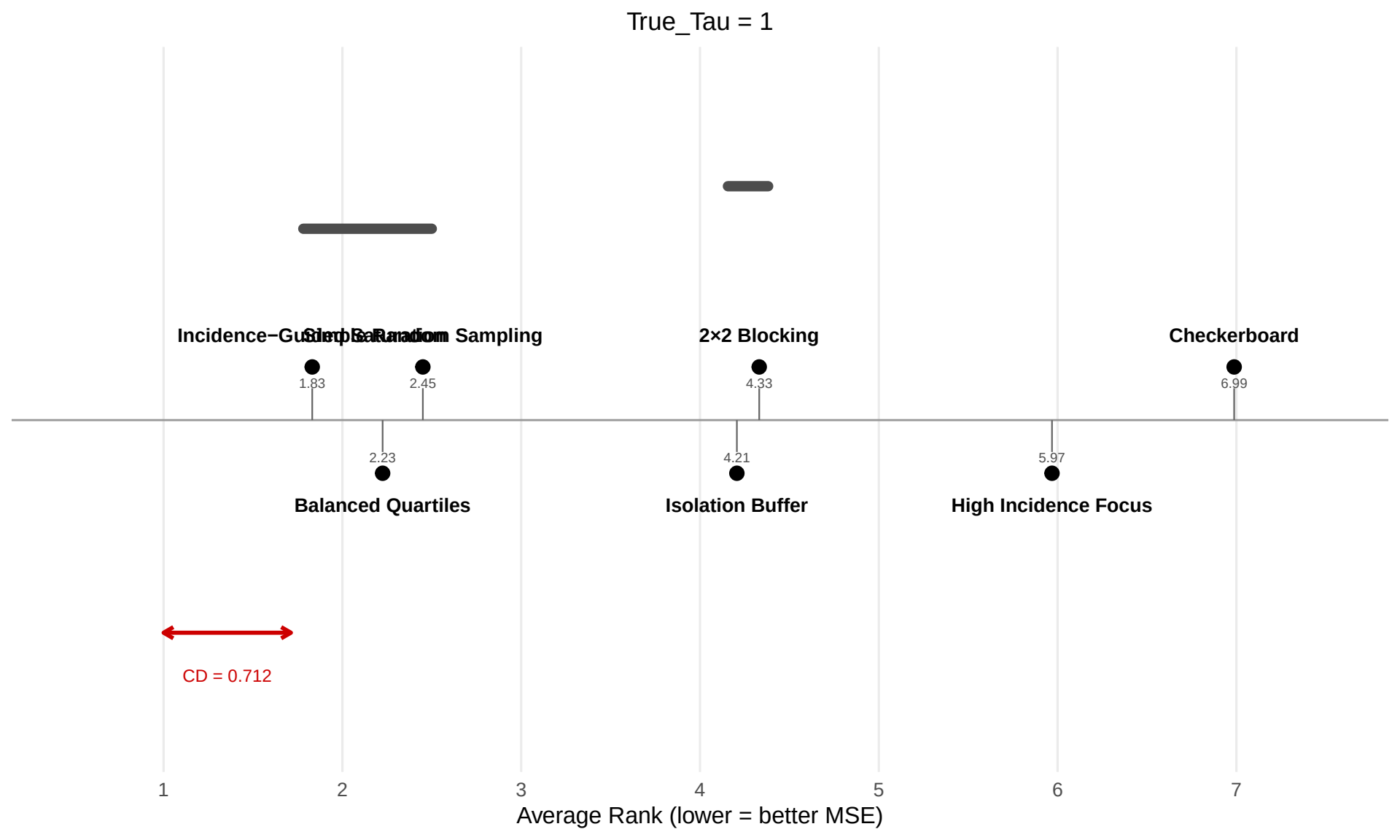

Critical Difference Diagram (6 designs + SRS, rook, tau=1.0)
Incidence−Guided Sampling
Simple Random Sampling
2×2 Blocking
Checkerboard
1.83
2.45
4.33
6.99
2.23
4.21
5.97
Balanced Quartiles
Isolation Buffer
High Incidence Focus
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)

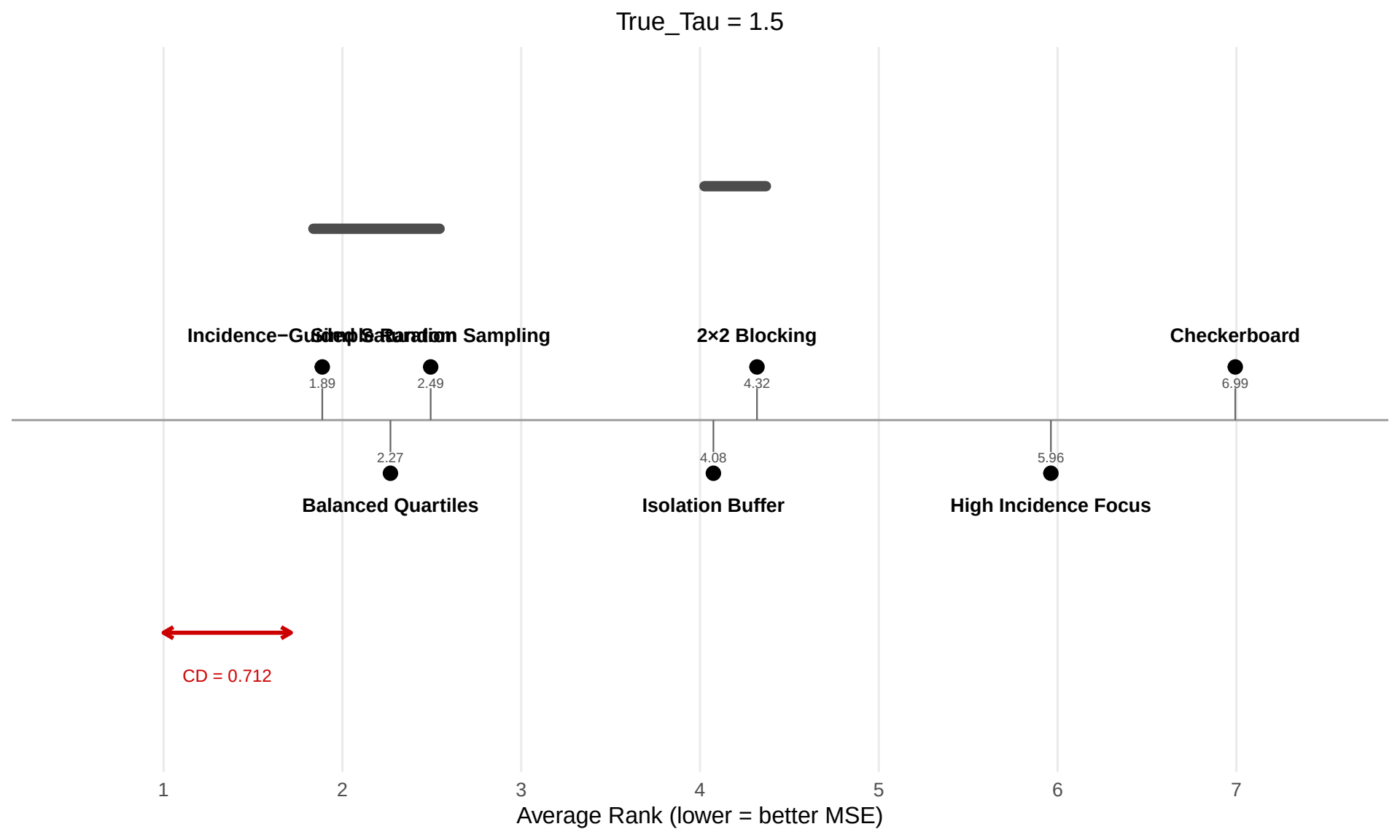

MSE by Design with Pairwise Significance
MSE
0.0
0.3
0.6
0.9
***
***
***
***
***
***
***
ncidence−Guided Saturation
Simple Random Sampling
Balanced Quartiles
2×2 Blocking
Isolation Buffer
High Incidence Focus
Checkerboard

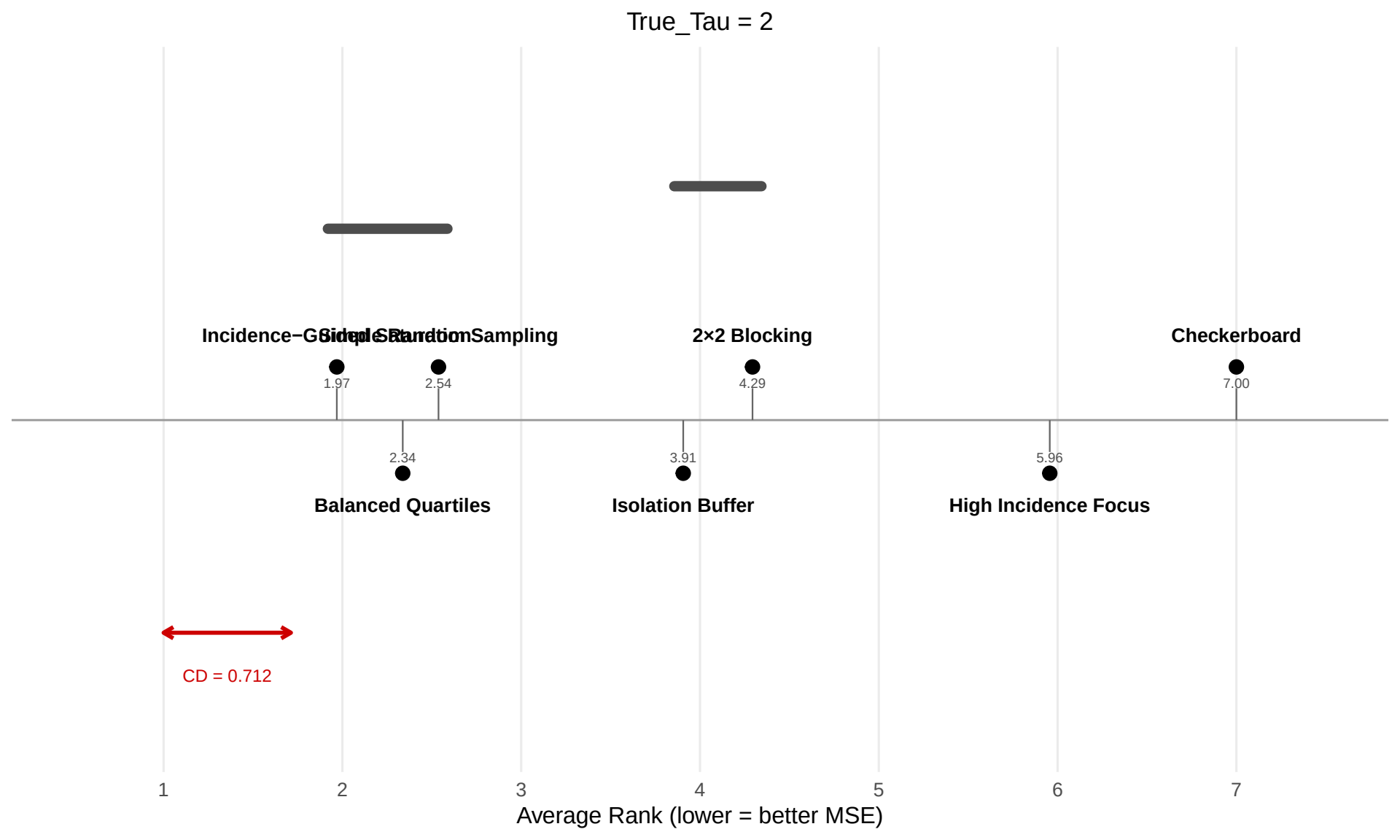
Pairwise Wilcoxon (Holm) p−values (Holm−adjusted)
2×2 Blocking
Balanced Quartiles
Checkerboard
High Incidence Focus
Incidence−Guided Saturation
Isolation Buffer
Simple Random Sampling
***
ns
p−value
1.00
0.50
0.10
0.001

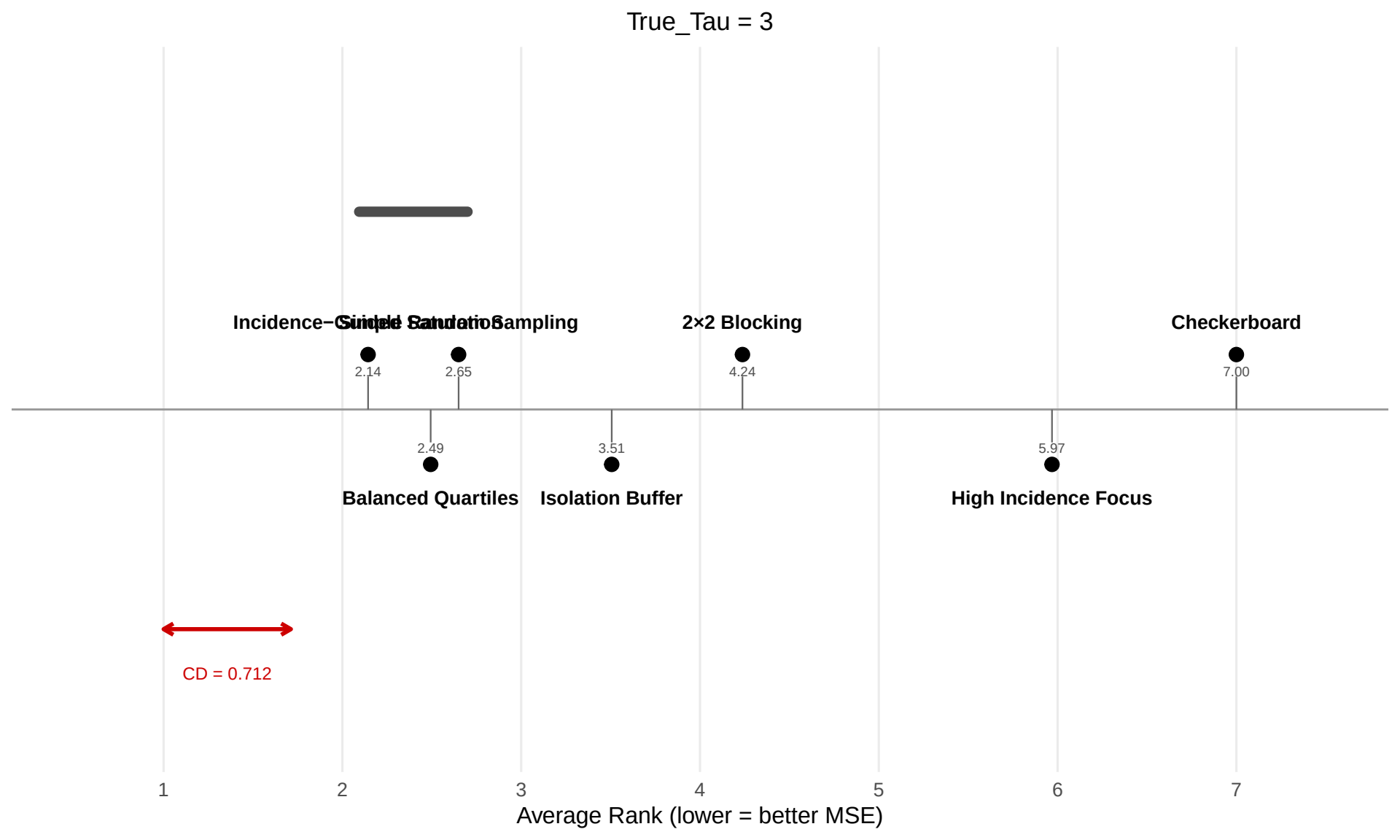

Pairwise Nemenyi p−values
2×2 Blocking
Balanced Quartiles
Checkerboard
High Incidence Focus
Incidence−Guided Saturation
Isolation Buffer
Simple Random Sampling
***
ns
p−value
1.00
0.50
0.10
0.001


True_Tau = 0.8
Incidence−Guided Stratified Sampling
Simple Random Sampling
2×2 Blocking
Checkerboard
1.81
2.42
4.33
6.99
2.22
4.26
5.97
Balanced Quartiles
Isolation Buffer
High Incidence Focus
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 1
Incidence−Guided Random Sampling
Simple Random Sampling
2×2 Blocking
Checkerboard
1.83
2.45
4.33
6.99
2.23
4.21
5.97
Balanced Quartiles
Isolation Buffer
High Incidence Focus
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 1.5
Incidence−Guided Separation
Simple Random Sampling
2×2 Blocking
Checkerboard
1.89
2.49
4.32
6.99
2.27
4.08
5.96
Balanced Quartiles
Isolation Buffer
High Incidence Focus
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 2
Incidence−Stratified Sampling
Simple Random
2×2 Blocking
Checkerboard
1.97
2.54
4.29
7.00
2.34
3.91
5.96
Balanced Quartiles
Isolation Buffer
High Incidence Focus
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 3
Incidence−Stratified Sampling
2.14
2.65
2×2 Blocking
4.24
Checkerboard
7.00
2.49
3.51
5.97
Balanced Quartiles
Isolation Buffer
High Incidence Focus
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)